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(19) **United States**(12) **Patent Application Publication**
Manalo(10) **Pub. No.: US 2013/0045737 A1**(43) **Pub. Date: Feb. 21, 2013**(54) **METHOD FOR INSTANT REGISTRATION OF
A ROAMING UE ONTO A PREFERRED
VPLMN FROM A NON-PREFERRED VPLMN****Publication Classification**(51) **Int. Cl.**
H04W 8/06 (2009.01)
H04W 64/00 (2009.01)(52) **U.S. Cl.** **455/433**(57) **ABSTRACT**

A User Equipment (UE) that was registered on a non-preferred network operator may be forced to register immediately and seamlessly onto another available more preferred network operator by updating the EF_LOCI in the UICC with the MCCMNC (i.e. PLMN ID) of the preferred network operator and then sending the REFRESH—3G Session Reset command to the Mobile Equipment (ME).

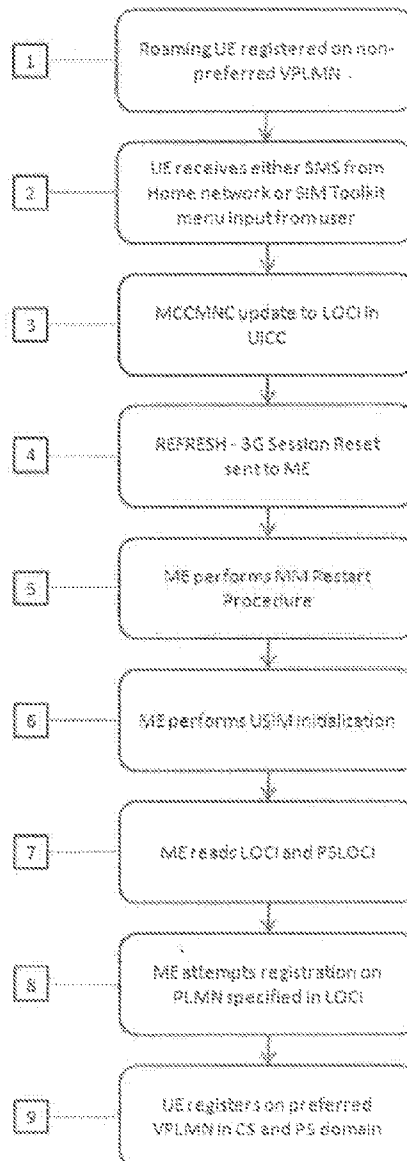
(75) Inventor: **Robert E. Manalo**, North York (CA)(73) Assignee: **TELUS COMMUNICATIONS
COMPANY**, Edmonton (CA)(21) Appl. No.: **13/212,132**(22) Filed: **Aug. 17, 2011****Method of Invention Flow Diagram**

Figure 1: Method of Invention Flow Diagram

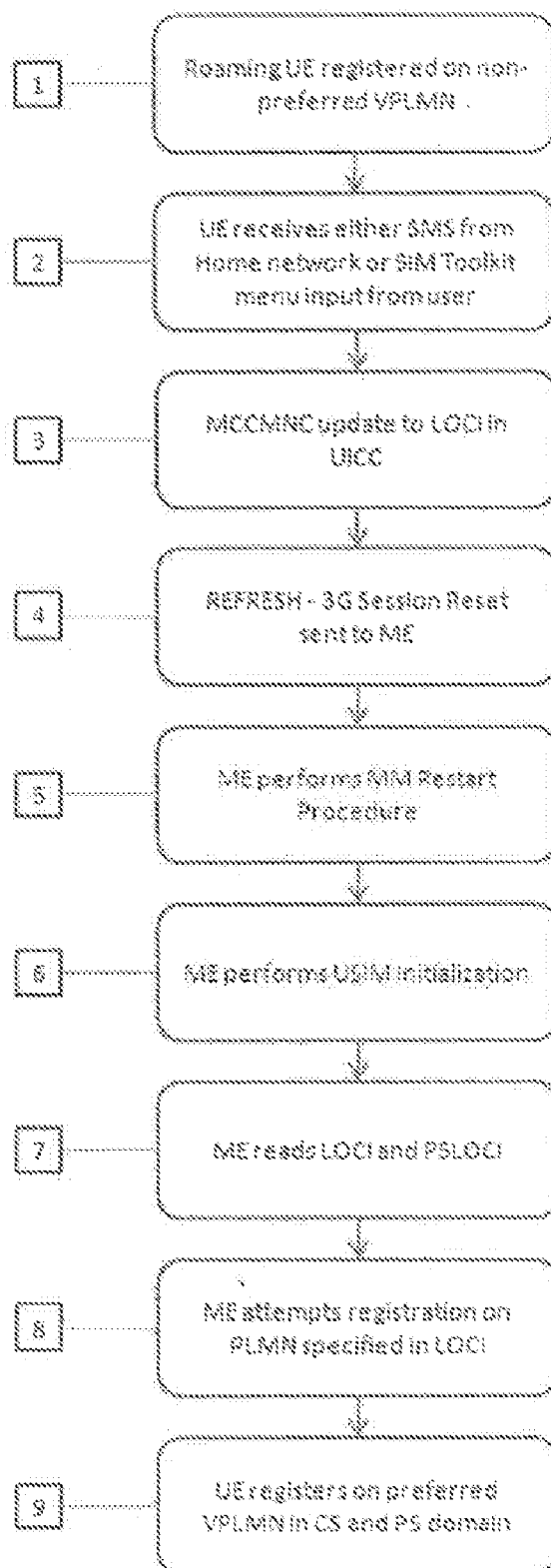


Figure 2: UE is powered ON

The screenshot displays a mobile application interface. On the left, a list of actions is shown, including (Cold Reset), SELECT (MF), SELECT (PL), READ BINARY (Offset 0, Len 10), SELECT (ICCID), READ BINARY (Offset 0, Len 10), TERMINAL PROFILE, SELECT (DIR), READ RECORD (1), SELECT (USIM ADF), SELECT (MF), SELECT (USIM ADF), SELECT (7F66), SELECT (USIM ADF), SELECT (1), READ BINARY (Offset 0, Len 10), SELECT (AD), READ BINARY (Offset 0, Len 4), VERIFY PIN (PIN Appl 1), VERIFY PIN (Second PIN Appl 1), UNBLOCK PIN (PIN Appl 1), UNBLOCK PIN (Second PIN Appl 1), SELECT (ECC), READ RECORD (1), READ RECORD (2), READ RECORD (3), and READ RECORD (4). On the right, a table with two columns, 'Name' and 'Value', is visible. The table contains two rows: 'Event' with the value 'Reset card interface' and 'Time' with the value '4.404 s'. At the bottom, a status bar shows 'Step 10 of 13' and a list of events: EVENT DOWNLOAD, SMS PP DOWNLOAD, SEND SHORT MESSAGE, REFRESH, EVENT DOWNLOAD, and EVENT DOWNLOAD. Below the status bar, there are tabs for 'Proactive session[s]' and 'Terminal Profile'.

Name	Value
Event	Reset card interface
Time	4.404 s

Step 10 of 13

EVENT DOWNLOAD: [icon] SMS PP DOWNLOAD [icon] SEND SHORT MESSAGE [icon] REFRESH [icon] EVENT DOWNLOAD: [icon] EVENT DOWNLOAD: [icon]

Session(s) abstract

ME -> USCC

Event: Location status

Location Status: No service

Proactive session[s] Terminal Profile

Figure 3: ME reads last registered PLMN in EF_LOCI

The screenshot displays a mobile terminal interface with a list of commands on the left and a detailed view of the selected command on the right. The commands include various SELECT, READ RECORD, and READ BINARY operations. The selected command, 'READ BINARY (Offset 0, Len 11)', shows a 'Name' column with fields like Command, Status, Current DF, Current EF, Time, Offset, Length, Logical Channel, and Secure Messaging Indication, and a 'Value' column with corresponding data.

Name	Value
Command	READ BINARY
Status	Normal ending of command
Current DF	USIM Application Directory (3F00.7FFF)
Current EF	Location Information (6F7E)
Time	27.682 s
Offset	0
Length	11
Logical Channel	90
Secure Messaging Indication	No SM used between terminal and card

Below the command list, there is a section for 'TMSI-Temporary Mobile Subscriber Identity' with the value '74 10 A6 20'. This is followed by 'LAI-Location Area Information' with the value 'MCC 302, MNC 880, LAC APC8'. The 'RFU or TMSI TIME' is 'FF', and the 'only for Phase 1 MEs' field is empty. The 'Location update status' is 'RFU bits 8-4: 00000' and 'Location update status: Updated'.

At the bottom, there is a 'Proactive session[s]' section with a 'Terminal Profile' tab. The 'Session(s) abstract' shows 'ME -> UBCC' and 'Event: Location status: No service'.

Figure 4: ME reads last registered PLMN in EF_PSLOC1

The screenshot displays a mobile terminal emulator interface. On the left, a list of commands is shown, including SELECT (MF), SELECT (USIM ADF), SELECT (OPL), READ RECORD (7), SELECT (APR), READ RECORD (9), SELECT (LOC1), READ BINARY (Offset 0, Len 11), SELECT (MF), SELECT (USIM ADF), SELECT (OPL), READ RECORD (8), SELECT (APR), READ RECORD (10), SELECT (PSLOC1), READ BINARY (Offset 0, Len 14), SELECT (MF), SELECT (USIM ADF), SELECT (OPL), READ RECORD (9), SELECT (APR), READ RECORD (11), SELECT (MF), READ BINARY (Offset 0, Len 1), SELECT (MF), SELECT (USIM ADF), and SELECT (OPL). The right pane shows the details for the selected 'READ BINARY' command. The details are as follows:

Name	Value
Command	READ BINARY
Status	Normal ending of command
Current EF	USIM Application Directory (3F00 7FFF)
Current EF	Packet Switched location information (6F73)
Time	27.979 s
Offset	0
Length	14
Logical Channel	00
Secure Messaging Indication	No SM used between terminal and card
P-TMSI	F4 42 51 52
P-TMSI signature	FF FF FF
RAI	MCC 302 MNC 000 LAC APC3 RAC FA
Routing Area update status	
RAI bits 0-4	00000
Routing Area update status	Updated

At the bottom of the emulator, there is a 'Proactive session[s]' tab and a 'Terminal Profile' tab. The 'Proactive session[s]' tab is active, showing a list of sessions. The 'Terminal Profile' tab is also visible.

Figure 5: ME reads if any PLMNs are listed in EF_OPLMNwAct list

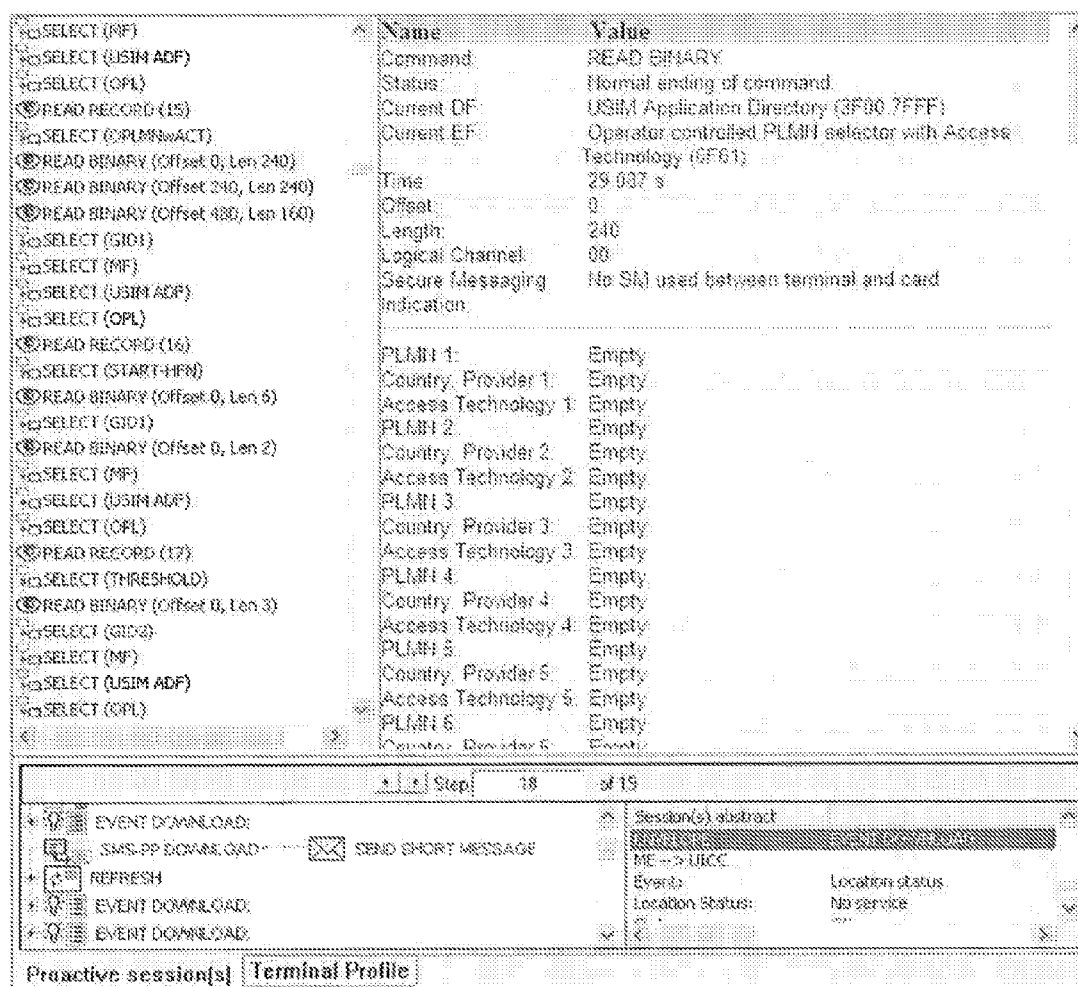


Figure 6: ME performs CS Authentication on last registered PLMN

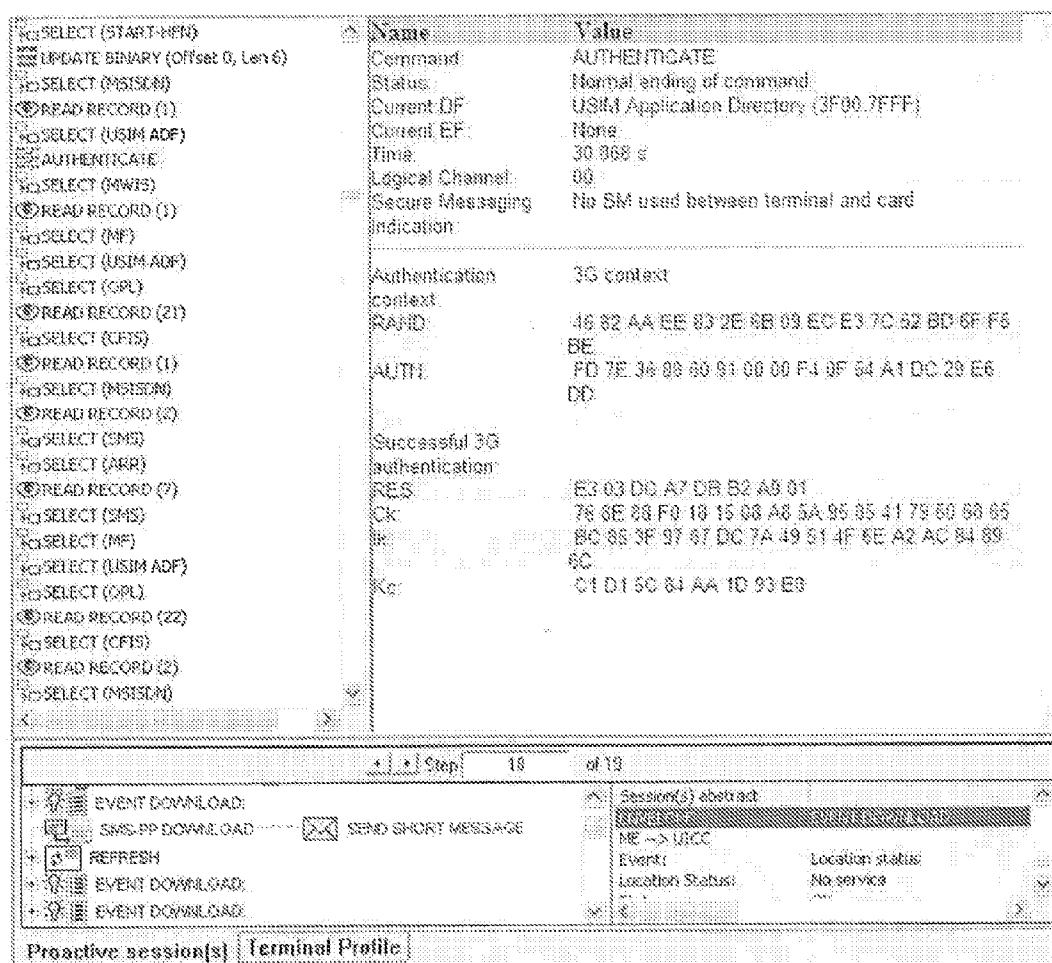


Figure 7: CS registration on last registered PLMN is successful

Name	Value
Command	UPDATE BINARY
Status	Normal ending of command
Current DF	USIM Application Directory (3F00.7FFF)
Current EF	Location Information (6F7E)
Time	31.892 s
Offset	0
Length	11
Logical Channel	60
Secure Messaging Indication	No SIM used between terminal and card
TMSI-Temporary	74 11 19 28
Mobile Subscriber Identity	
LAI-Location Area Information	MCC 302 MNC 890 LAC AFCS
RFU or TMSI TIME (only for Phase 1 MEs)	FF
Location update status	
RFU bits 8-4	00000
Location update status	Updated

Step 18 of 19	
EVENT DOWNLOAD	Session(s) abstract
SMS-PP DOWNLOAD	ME -> UICC
SEND SHORT MESSAGE	Event: Location status
REFRESH	Location status: No service
EVENT DOWNLOAD	
EVENT DOWNLOAD	

Proactive session[s] Terminal Profile

Figure 8: Normal Service on non-preferred 302-880 VPLMN in CS Domain

The screenshot displays a mobile device interface with a list of commands on the left and a detailed view of the selected command on the right.

Command List (Left):

- SELECT (MF)
- SELECT (USIM ADF)
- SELECT (OPL)
- READ RECORD (25)
- SELECT (CFIS)
- READ RECORD (3)
- SELECT (LOC)
- UPDATE BINARY (Offset 0, Len 11)
- ENVELOPE (EVENT DOWNLOAD)**
- SELECT (SMSP)
- SELECT (SMS)
- READ RECORD (1)
- SELECT (USIM ADF)
- AUTHENTICATE
- SELECT (MF)
- SELECT (USIM ADF)
- SELECT (OPL)
- READ RECORD (24)
- SELECT (FC)
- UPDATE BINARY (Offset 0, Len 9)
- SELECT (Keys)
- UPDATE BINARY (Offset 0, Len 33)
- SELECT (CFIS)
- READ RECORD (4)
- SELECT (SMSP)
- READ RECORD (1)
- SELECT (SMS)

ENVELOPE (EVENT DOWNLOAD) Details (Right):

Name	Value
Command	ENVELOPE
Status	Normal ending of command
Current CF	USIM Application Directory (3F00 7FFF)
Current EF	Location Information (6F7E)
Time	31 936 s
Logical Channel	00
Secure Messaging	No SM used between terminal and card
Indication	
EVENT DOWNLOAD	
Event List	
Event	Location status
Device Identity	
Source	ME
Destination	UICC
Location Status	
Contents	Normal service
Location Information	
Mobile Country	302
Codes (MCC)	
Mobile Network	980
Codes (MNC)	
Location Area Code	AFC8
(LAC)	
Cell Identity Value	732B
Cell ID	
Extended Cell ID	06 DD
(hex)	

Bottom Section:

Step 5 of 19

SMS-PP DOWNLOAD ☒ SEND SHORT MESSAGE

REFRESH

EVENT DOWNLOAD

EVENT DOWNLOAD

Session(s) abstract

ME -> UICC

Event: Location status

Location Status: Normal service

Proactive session(s) Terminal Profile

Figure 9: ME performs PS Authentication on last registered PLMN

The screenshot displays a mobile terminal interface with a list of commands on the left and a detailed view of the 'AUTHENTICATE' command on the right. The 'AUTHENTICATE' command shows the following details:

Name	Value
Command	AUTHENTICATE
Status	Normal ending of command
Current DF	USIM Application Directory (3F00 7FFF)
Current EF	None
Time	32.358 s
Logical Channel	00
Secure Messaging Indication	No SM used between terminal and card
Authentication context	3G context
RAND	96 6B E9 6D 8E 8C 3E 14 EA 16 8D 0A 34 CA DE 3D
AUTH	3B 6F EC 2E 32 FB 00 00 34 6A 09 E1 8E A9 73 CF
Successful 3G authentication	
RES	87 64 42 B4 A3 0E 1E D3
CK	28 39 FE C6 7A B3 01 80 97 66 6A B7 A4 3A C4 E4
IK	FE 06 07 6A 96 EC E5 62 DA D6 00 74 74 39 48 60
Kc	0B 8C 93 6F 3C 8C 69 6F

Below the command details, there is a section for 'Proactive session(s)' and 'Terminal Profile'. The 'Proactive session(s)' section shows a list of sessions, including 'GSM-PP DOWNLOAD' and 'SEND SHORT MESSAGE'. The 'Terminal Profile' section shows the 'Session(s) abstract' and 'ME -> UICC' status, with 'Location status' set to 'Normal service'.

Figure 10: PS registration on last registered PLMN is successful

The screenshot displays a mobile terminal interface with a list of commands on the left and a detailed view of the 'UPDATE BINARY' command on the right. The bottom section shows a 'Session(s) abstract' and a 'Terminal Profile'.

Name	Value
Command	UPDATE BINARY
Status	Normal ending of command.
Current DF	USIM Application Directory (3F00 7FFF)
Current EF	Packet Switched location information (6F73)
Time	33.426 s.
Offset	0
Length	14
Logical Channel	00
Secure Messaging Indication	No SM used between terminal and card
P-TMSI	F4 42 03 A2
P-TMSI signature	C7 E9 E1
RAI	MCC 302 MNC 830 LAC AF0B RAC FA
Routing Area update status	
RFU bits 8-4	00000
Routing Area update status	Updated

Session(s) abstract

ME → UICC

Event: Location status

Location Status: Normal service

Proactive session[s] Terminal Profile

Figure 11: UE receives SMS from Home Network

Name	Value
Command	ENVELOPE
Status	Normal ending of command, with extra information from the proactive UICC. Length 51
Current DF	USIM Application Directory (2F00 7FFF)
Current EF	Initialisation values for Hyperlane number (6F6B)
Time	92.242 s
Logical Channel	00
Secure Messaging Indication	No SM used between terminal and card
SMS-PP DOWNLOAD	
Device Identity	
Source	Network
Destination	UICC
Address	
Type of number	International Number
Numbering plan	ISDN/ telephony numbering plan (Rec. E 164)
Identification	E 163
Dialling number	16472138204
string	
SMS TPDU	
TP-Message-Type	SMS-DELIVER (in the direction SC to MS)
Indicator	
TP-Acta-	No more messages are waiting for the MS in this
Messages-to-Send	SC
TP-Reply-Path	TP-Reply-Path parameter is not set in this SMS-
	SUBMIT/DELIVER
TP-User-Data-	The beginning of the TP-UD field contains a Header
Header indicator	in addition to the short message
Step 6 of 13	
Session(s) abstract	
Network--> UICC	
Address (Dialing Numb... +16472138204	
TP-Originating-Address 3520	
Proactive session(s) Terminal Profile	

Figure 12: REFRESH – 3G Session Reset is sent to the ME

The screenshot displays a software interface for managing commands. On the left, a list of commands is shown, including 'STATUS (USIM ADF)', 'UPDATE BINARY (Offset 0, Len 6)', 'ENVELOPE (SMS-PP DOWNLOAD)', 'FETCH (SEND SHORT MESSAGE)', 'SELECT (Kc)', 'UPDATE BINARY (Offset 0, Len 9)', 'SELECT (Keys)', 'UPDATE BINARY (Offset 0, Len 33)', 'SELECT (START-HFN)', 'UPDATE BINARY (Offset 0, Len 6)', 'SELECT (SMSS)', 'READ BINARY (Offset 0, Len 2)', 'UPDATE BINARY (Offset 0, Len 2)', 'SELECT (START-HFN)', 'UPDATE BINARY (Offset 0, Len 6)', 'TERMINAL RESPONSE (SEND SHORT MES', 'FETCH (REFRESH)', 'SELECT (LI)', 'READ BINARY (Offset 0, Len 10)', 'SELECT (AD)', 'READ BINARY (Offset 0, Len 4)', 'SELECT (ECC)', 'READ RECORD (1)', 'READ RECORD (2)', 'READ RECORD (3)', 'READ RECORD (4)', and 'READ RECORD (5)'. The 'REFRESH' command is selected, and its details are shown on the right.

Name	Value
Command	FETCH
Status	Normal ending of command
Current DF	USIM Application Directory (3F00.7FFF)
Current EF	Initialisation values for Hyperframe number (6F5B)
Time	97.453 s
Logical Channel	00
Secure Messaging Indication	No SM used between terminal and card
Proactive UICC	
Command	
Command Details	
Command Number	2
Command Name	REFRESH
Command Qualifier	3G Session Reset
Device Identity	
Source	UICC
Destination	ME
File List	
Number of files	1
File	3F007FFF6F7E

Proactive session(s) | Terminal Profile

Session(s) abstract
UICC -> ME
File: 3F007FFF6F7E

Figure 13: ME re-reads EF_LOCI containing updated MCCMNC during USIM Initialization process

The screenshot displays a mobile terminal interface during USIM initialization. The top section shows a list of commands on the left and their corresponding values on the right. The commands include SELECT (MF), SELECT (USIM ADF), SELECT (OPL), READ RECORD (15), READ BINARY (Offset 0, Len 11), READ BINARY (Offset 0, Len 14), READ RECORD (17), READ BINARY (Offset 5, Len 1), READ RECORD (18), READ BINARY (Offset 0, Len 9), and SELECT (OPL). The values include Command (READ BINARY), Status (Normal ending of command), Current DF (USIM Application Directory (3F00 7FFF)), Current EF (Location Information (8F7E)), Time (101 085 s), Offset (0), Length (11), Logical Channel (00), Secure Messaging Indication (No SM used between terminal and card), TMSI-Temporary Mobile Subscriber Identity (74 11 19 28), LAI-Location Area Information (MCC 302 MNC 221 LAC AFC8), RFU or TMSI TIME (FP), Location update status (RFU bits 0-4: 00000), and Location update status (Updated).

The bottom section shows a progress bar at Step 11 of 19. Below the progress bar, there are several checkboxes: EVENT DOWNLOAD, SMS PP DOWNLOAD, SEND SHORT MESSAGE, REFRESH, and EVENT DOWNLOAD. The right side of the bottom section shows a list of session(s) abstract, with the first entry being ME -> UICC. Below this, there are fields for Event (Location status) and Location status (No service).

At the bottom of the screen, there are two tabs: Proactive session(s) and Terminal Profile.

Figure 14: UE re-reads PSLOC containing updated MCCMNC during USIM Initialization process

The screenshot displays a mobile device interface with a list of USIM initialization commands on the left and a detailed log of the 'READ BINARY' command on the right. Below these, a status bar shows 'Step 11 of 19' and a 'Session(s) abstract' window displays 'Location status: No service'.

Name	Value
Command	READ BINARY
Status	Normal ending of command
Current DF	USIM Application Directory (3F00 7FFF)
Current EF	Packet Switched location information (6F73)
Time	101.241 s
Offset	0
Length	14
Logical Channel	00
Secure Messaging Indication	No SM used between terminal and card
P-TMSI	FA 42 63 A2
P-TMSI signature value	FF FF FF
RAI	MCC 302 MNC 880 LAC AFC0 RAC FA
Routing Area update status	RFU bits 0-4 00000
Routing Area update status	Updated

Step 11 of 19

Session(s) abstract

Location status: No service

Figure 15: ME performs REFRESH – 3G Session Reset successfully

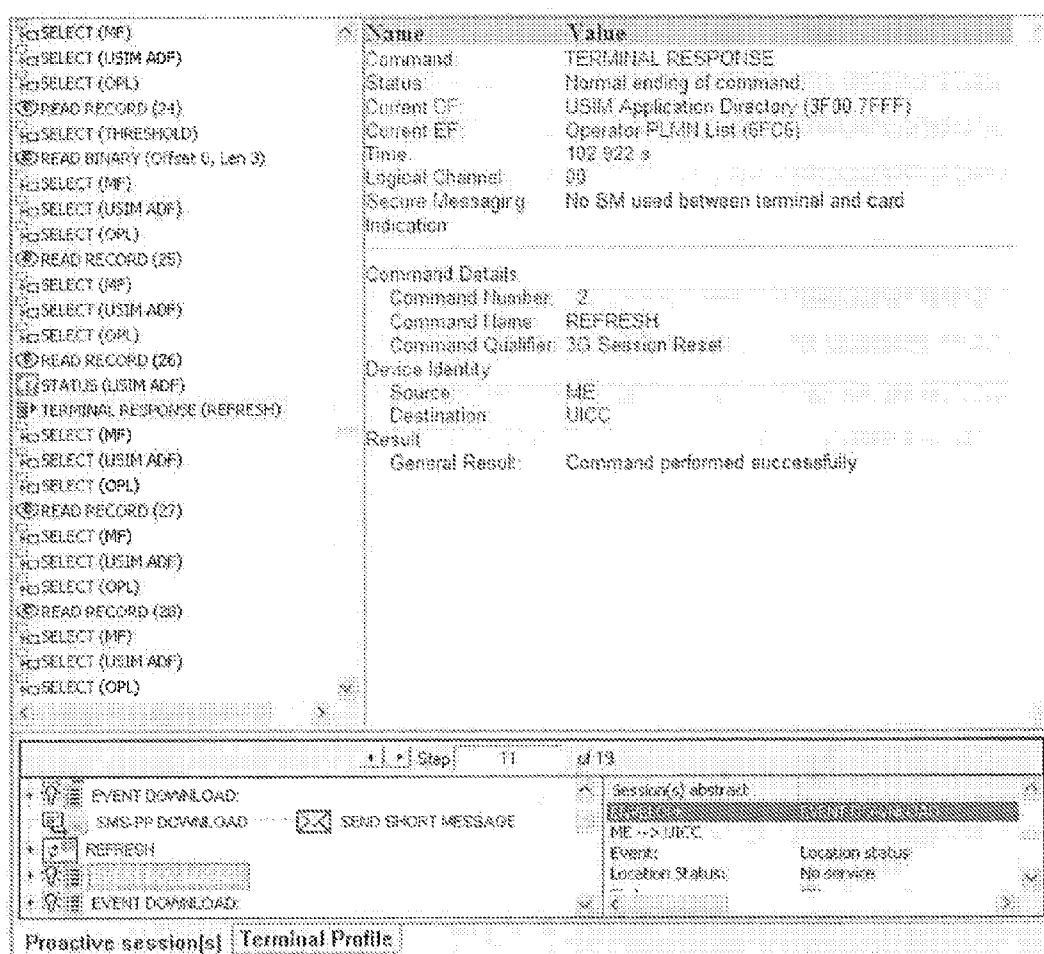


Figure 16: ME performs CS Authentication on updated last registered PLMN

The screenshot displays a mobile device interface with a list of commands on the left and a detailed view of the AUTHENTICATE command on the right.

Command List (Left):

- SELECT (USIM ADF)
- AUTHENTICATE
- SELECT (MF)
- SELECT (USIM ADF)
- SELECT (OPL)
- READ RECORD (19)
- SELECT (USIM ADF)
- AUTHENTICATE
- SELECT (MF)
- SELECT (USIM ADF)
- SELECT (OPL)
- READ RECORD (80)
- SELECT (KcSPRS)
- UPDATE BINARY (Offset 0, Len 9)
- SELECT (KeysPS)
- UPDATE BINARY (Offset 0, Len 35)
- SELECT (MF)
- SELECT (USIM ADF)
- SELECT (PIN)
- SELECT (PSLOC)
- UPDATE BINARY (Offset 0, Len 14)
- SELECT (NETPAR)
- UPDATE BINARY (Offset 0, Len 218)
- SELECT (LOC)
- UPDATE BINARY (Offset 0, Len 11)
- ENVELOPE (EVENT DOWNLOAD)
- SELECT (Kc)

AUTHENTICATE Command Details (Right):

Name	Value
Command	AUTHENTICATE
Status	Normal ending of command
Current DF	USIM Application Directory (3F00.7FFF)
Current EF	None
Time	105.545 s
Logical Channel	00
Secure Messaging Indication	No SM used between terminal and card
Authentication context	3G context
RAND	33 F3 8D E6 A2 D0 66 6B 1D A8 92 FE 20 C0 F5 7C
AUTH	3D E0 F2 C2 8E 1C 00 00 18 E1 26 10 0F 7F 37 68
Successful 3G authentication	
RES	F1 40 25 E3 2E 68 3E B0
CK	68 49 0B 09 65 05 06 06 06 A0 45 C0 07 E4 EE 3F
IK	BC F5 18 AC 54 10 A4 B5 3F 18 1F 06 E2 60 D9 60
Kc	70 04 49 63 D4 91 95 2C

Bottom Panel:

Step 11 of 19

EVENT DOWNLOAD: SMS-PP DOWNLOAD SEND SHORT MESSAGE

REFRESH

EVENT DOWNLOAD

Proactive session(s) Terminal Profile

Session(s) abstract

ME -> LICC

Event: Location status

Location Status: No service

Figure 17: ME performs PS Authentication

Name	Value
Command:	AUTHENTICATE
Status:	Normal ending of command
Current DF:	USIM Application Directory (3F00.7FFF)
Current EF:	None
Time:	105.872 s
Logical Channel:	00
Secure Messaging Indication:	No SM used between terminal and card
Authentication context:	3G context
RAND:	71 44 C6 03 D2 5E 99 BE FA 61 4F 87 EE 07 ED FB
AUTH:	2E 12 7D 5A F0 C7 00 00 37 0B F2 E2 D9 3E 87 22
Successful 3G authentication:	
RES:	2F B9 33 75 7B 0F 02 A2
CK:	50 0D F5 06 25 93 89 84 D6 19 D2 6B E1 FE 2B 6E
Kc:	6E E1 DE 8A 31 1C B3 EC C0 06 31 14 CD CC 27 2B
Kc:	28 F5 C8 4D 38 6D 3F 2D

Proactive session[s] Terminal Profile

Figure 18: PS registration on updated last registered PLMN is successful

The screenshot displays a mobile device interface with a list of commands on the left and a detailed view of the 'UPDATE BINARY' command on the right. The bottom section shows session management options and a terminal profile.

Name	Value
Command	UPDATE BINARY
Status	Normal ending of command
Current DF	USIM Application Directory (3F00 7FFF)
Current EF	Packet Switched location information (6F73)
Time	107.303 s
Offset	0
Length	14
Logical Channel	00
Secure Messaging Indication	No SM used between terminal and card
P-TMSI	CA 28 02 67
P-TMSI signature	CE 64 69
RAI	MCC 302 . MNC 221 . LAC 000B . RAC 0B
Routing Area update status	RFU bits 0-4: 00000
Routing Area update status	Updated

Session(s) abstract:

Session	State
1	ACTIVE

ME --> UICC

Events: Location status: No service

Location Status: No service

Proactive session(s) | Terminal Profile

Figure 19: CS registration on updated last registered PLMN is successful

Name	Value
Command	UPDATE BINARY
Status	Normal ending of command
Current DF	USIM Application Directory (3F00 7FFF)
Current EF	Location Information (6F7E)
Time	107.503 s
Offset	0
Length	11
Logical Channel	00
Secure Messaging	No SM used between terminal and card
Indication	
TMSI-Temporary	07 41 FA 3F
Mobile Subscriber	
Identity	
LAI Location Area	MCC 392 MNC 291 LAC 8008
Information	
RFU or TMSI TIME	FF
only for Phase 1	
MEs:	
Location update	
status:	
RFU bits 8-4	00000
Location update	Updated
status:	

Step 11 of 19	
EVENT DOWNLOAD ☐ SMS-PP DOWNLOAD ☒ SEND SHORT MESSAGE 2 REFRESH EVENT DOWNLOAD	Session(s) abstract ME -> UICC Event: Location status Location Status: No service

Proactive session(s) Terminal Profile

Figure 20: UE is fully registered on more preferred 302-221 VPLMN in ~10 seconds after REFRESH

The screenshot displays a mobile terminal interface with a list of commands on the left and a detailed view of the 'ENVELOPE' command on the right. Below these, a 'Proactive session(s)' window shows the 'EVENT DOWNLOAD' and 'REFRESH' commands, along with a 'Terminal Profile' section.

Name	Value
Command	ENVELOPE
Status	Normal ending of command
Current DF	USIM Application Directory (3F00 7FFF)
Current EF	Location Information (6F7E)
Time	107.534 s
Logical Channel	00
Secure Messaging Indication	No SIM used between terminal and card
EVENT DOWNLOAD	
Event List	
Event	Location status
Device Identity	
Source	ME
Destination	UICC
Location Status	
Contents	Normal service
Location Information	
Mobile Country	302
Codes (MCC)	
Mobile Network	221
Codes (MNC)	
Location Area Code	0008
(LAC)	
Cell Identity Value	2973
(Cell ID)	
Extended Cell ID	00 0B
(hex)	

Proactive session(s) **Terminal Profile**

Step: 12 of 13

Session(s) abstract

EVENT DOWNLOAD

ME -> UICC

Event: Location status

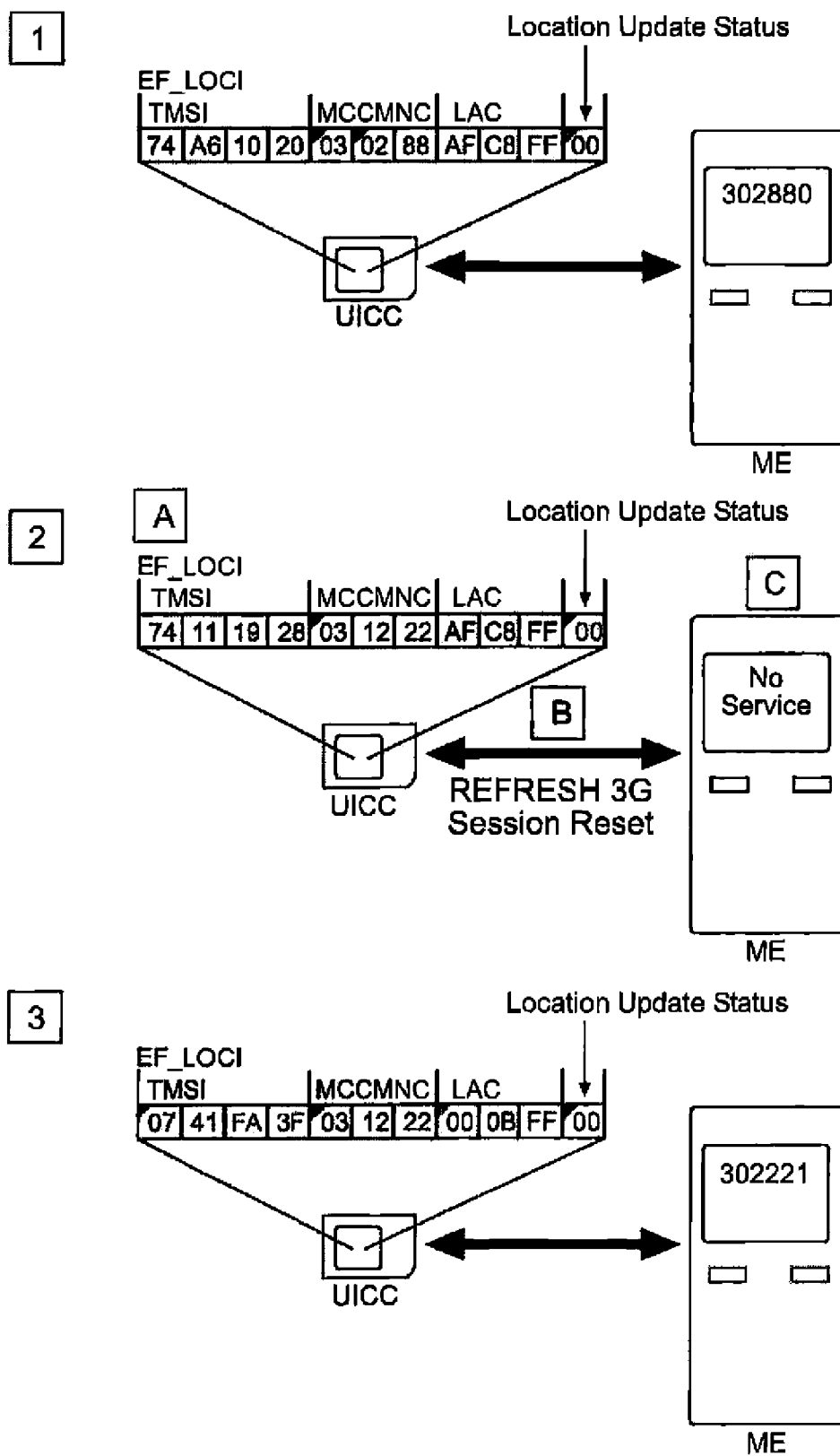
Location Status: Normal service

REFRESH

EVENT DOWNLOAD

SEND SHORT MESSAGE

Figure 21: UICC and ME interaction



METHOD FOR INSTANT REGISTRATION OF A ROAMING UE ONTO A PREFERRED VPLMN FROM A NON-PREFERRED VPLMN

FIELD

[0001] Network Selection Procedure of a Mobile Equipment through interaction with updates to and commands from the UICC.

BACKGROUND AND SUMMARY

[0002] The disclosed method is an innovative new seamless instant registration method by which a roaming UE that is in Automatic Network Selection mode and registered on a non-preferred visited network, can be forced to immediately register seamlessly within seconds onto another available more preferred visited network. The general method by which this can be accomplished is by updating a specific file in the UICC with specific content, followed by a sending a specific REFRESH command to the ME. This method can be deployed using either a commercial OTA solution or a SIM Toolkit Application in the UICC.

[0003] When a roaming UE is powered on in a foreign country with multiple visited networks available, for which registration is possible (due to bilateral network agreements), if the ME is set to Automatic Network Selection mode (which typically is the default setting), the ME follows the Network Selection Procedures outlined in 3GPP TS 23.122 clause 4.4.3.1 and 4.4.3.3.

[0004] There are some scenarios where if the OPLMN-NwAcT preferred roaming list in the UICC is empty or not updated with the latest more preferred network, then the roaming UE may register onto a less preferred visited network that the home operator would not want their client's UE to register onto. Reasons may be higher roaming costs and/or fewer services such as 3G data offered on the less preferred network.

[0005] In these scenarios, the following United States patent documents generally describe an OTA network based solution in which to steer a client's UE (i.e. via OTA updates to the LOCI and PLMN list followed by a REFRESH) to a more preferred visited network: U.S. Pat. No. 7,127,245 Almgren and U.S. Pat. No. 7,139,570 Elkarat et al.

[0006] However, the specific method for how to accomplish the seamless and instant registration behaviour of the UE onto the more preferred visited network after the OTA update is not covered in these patents, for which the descriptions and claims presented in these Prior Arts are obscure and ambiguous in this regard and leave the reader who is skilled in this subject matter area to spend countless hours experimenting and testing with different REFRESH command combinations with different combinations of LOCI updates.

[0007] Described are some examples of the challenges and limitations of the at least some previous methods. First, if the LOCI file is updated with the PLMN ID of the preferred visited network, but the Location update status byte 11 is set to a value other than "00", then seamless and instant registration will not work for all MEs, regardless of which REFRESH command is used after the LOCI file is updated. Second, if the LOCI file is updated with the PLMN ID of the preferred visited network, but the TMSI value of "FF FF FF FF" is specified in bytes 1 to 4, then seamless and instant registration will not work in some MEs, even if used with REFRESH—3G Session Reset. Third, even if the LOCI file is properly

updated by updating the PLMN ID of the preferred PLMN, while keeping the TMSI and Location update status intact with the previous values (i.e. non-"FF FF FF FF" TMSI value and Location update status set to "00"), usage of REFRESH—File Change Notification will not accomplish instant registration onto the preferred network on all MEs. Fourth, even if the LOCI file is properly updated by updating the PLMN ID of the preferred PLMN, while keeping the TMSI and Location update status intact with the previous values, usage of REFRESH—UICC Reset will not accomplish seamless and instant registration on MEs that implement a "Cold Reset" procedure, in which the ME performs a powercycle which will give the client the perception that the UE has malfunctioned. Fifth, only REFRESH—3G Session Reset specifies a re-initialization of a network connection through the MM Restart Procedure as per 3GPP TS 31.111 clause 6.4.7. As such, it is not guaranteed and required as per 3GPP standards (unless there is an IMSI change as per 3GPP TS 11.14 clause 6.4.7.1) that all MEs using other REFRESH commands, will perform a new network selection attempt on the preferred PLMN using the updated LOCI, even if the LOCI file is properly updated by updating the PLMN ID of the preferred PLMN, while keeping the TMSI and Location update status intact with the previous values.

[0008] Therefore, the disclosed method is a novel improvement to these methods, which focuses on only updating the MCCMNC content in the LOCI file with the preferred PLMN while keeping the TMSI and Location update status intact with the previous values, followed by a REFRESH—3G Session Reset command to the ME, in order to obtain immediate registration within seconds, for which there is no precedence.

[0009] Furthermore, the method for seamless instant registration through an MCCMNC update to the LOCI file followed by REFRESH—3G Session Reset to the ME, can also be accomplished through a SIM Toolkit application, in which perhaps the user, via a SIM Toolkit menu in the ME UI, selects the current visited country from a list of countries shown followed by selecting a predefined Network Name associated to the preferred PLMN code of the selected visited country. Once selected, the SIM Toolkit app proceeds with carrying out the disclosed method.

[0010] In conjunction to using the method for seamless instant registration onto the preferred visited network, the OPLMN-NwAcT file in the UICC can also be updated (i.e. using either OTA solution or SIM Toolkit) with the preferred visited network PLMN ID as higher priority over other PLMN IDs (if populated in the OPLMN-NwAcT list) of the same visited country, followed by sending the REFRESH—3G Session Reset command to the ME. This will help keep the UE registered on the preferred visited network, if available, during an HPLMN rescan.

[0011] In the future, a new REFRESH command requirement specified in 3GPP Release 7, REFRESH—Steering of Roaming, is aimed to provide similar instant and seamless registration behaviour as the disclosed method. However, REFRESH—Steering of Roaming is only applicable to Release 7 and higher MEs, and is not specified to work with today's current MEs, most of which are pre-Release 7. Therefore, the practicality of this new REFRESH command is not yet feasible.

BRIEF DESCRIPTION OF FIGS.

[0012] There will now be described embodiments of the disclosed method with reference to the drawings by way of example, in which:

[0013] FIG. 1 is a flow diagram showing an example of the method steps disclosed;

[0014] FIGS. 2-20 are screen shots illustrating the disclosed method steps; and

[0015] FIG. 21 is a schematic containing steps 2A, 2B, 2C and 3 that distinguish between method steps taken by the UICC and the ME.

DETAILED DESCRIPTION

[0016] The disclosed method leverages the expected behaviour of the ME as outlined in the following 3GPP requirements:

[0017] 3GPP TS 23.122 clause 4.4.3.1—At switch on or recovery from lack of coverage At switch on, or following recovery from lack of coverage, the MS selects the registered PLMN or equivalent PLMN (if it is available) using all access technologies that the MS is capable of and if necessary (in the case of recovery from lack of coverage, see clause 4.5.2) attempts to perform a Location Registration.

[0018] 3GPP TS 31.111 clause 6.4.7—REFRESH: 3G Session Reset. This mode causes the ME to reset the 3G session, in accordance with the 3G session reset procedure defined in TS 31.102 [14]. Subsequently, the ME performs the “USIM Initialization and File Change Notification” procedure and the MM Restart procedure as defined in TS 23.122 [7].

[0019] 3GPP TS 23.12.2 clause 4.8—MM Restart procedure: In some cases, e.g. on change of SIM data, there is a need for the MM to be restarted without the need for user intervention. To perform the procedure the MS shall behave as if the SIM is removed and afterwards a new SIM is inserted.

[0020] 3GPP TR 22.811 clause 5.4—Customer expectations: Following switch-on if the last registered PLMN is available, and no higher priority PLMN is available (e.g. HPLMN) the user should be registered on that PLMN (typically within 5 seconds). This period includes the execution of any 3GPP defined functionality as described above.

[0021] Based on the above 3GPP requirements, it can be Summarized that:

[0022] 1) The ME will first attempt the last registered PLMN.

[0023] 2) If the last registered PLMN is available and allowable, then the UE should register on that PLMN within 5 seconds.

[0024] 3) REFRESH—3G Session Reset will cause the ME to reset the 3G session, followed by USIM Initialization and File Change Notification, then performing an MM Restart (i.e. similar behaviour to removing and reinserting a new SIM).

[0025] Therefore, when a UE is registered on a non-preferred network, by overwriting the MCCMNC in the EF_LOCI file, and applying 3) above, the ME upon completing 3) will then proceed to 1) and attempt to register on the PLMN ID that was updated in the EF_LOCI, which the ME now considers as the last registered PLMN and completes 2), although registration will be within 30 seconds.

[0026] Note that when updating EF_LOCI with the PLMN ID of the preferred PLMN, depending on the GUI of the OTA Platform solution or SIM Toolkit Application, the user can either enter the PLMN ID of the preferred PLMN in decimal

value (i.e. 310410) which will be converted to raw hex format (i.e. 13 00 14) used by the UICC, or enter the PLMN ID directly in raw hex format.

[0027] Therefore, in an embodiment, there is disclosed a method of forcing a UE that is in Automatic Network Selection mode and is registered on a non-preferred visited network to register onto an available preferred visited network by the steps comprising overwriting an entry in an elementary file containing location information in the UE with identification for the preferred network, then sending a refresh command to the ME. In an embodiment, the method steps comprise overwriting the MCCMNC content of EF_LOCI (i.e. bytes 5, 6 and 7 of File ID 6F7E) in the USIM.ADF within the UICC, with the PLMN ID of an available preferred visited network, and then sending to the ME the refresh command REFRESH—3G Session Reset (containing EF_LOCI in the REFRESH File List).

[0028] FIG. 1 shows a flow diagram of the method of an embodiment of the disclosed method, for which the subsequent FIGS. 2 to 20, showing UICC-ME COMPRIION IT3 Prov! log file screen captures makes reference to.

[0029] FIG. 2 shows a screen shot of the UICC-ME log file where the UE is powered on and the ME begins reading the UICC files @ 4.404 s,

[0030] FIG. 3 shows the EF_LOCI first being read by the ME @ 27.682 s. The UE in this case was previously CS registered on PLMN 302-880, since the TMSI is populated and the Location update status is shown as “updated.”

[0031] FIG. 4 shows the EF_PSLOCI first being read by the ME @ 27.879 s. The UE in this case was previously PS registered on PLMN 302-880, since the P-TMSI is populated and the Routing Area update status is shown as “updated.”

[0032] FIG. 5 shows the EF_OPLMNwAcT being read by the ME @ 29.087 s. Although the screen shot does not fully show this EF in its entirety, the entries in this EF are all empty. This means that the ME will continue to use the last registered PLMN found in the EF_LOCI and EF_PSLOCI even during an HPLMN rescan, since there are no PLMNs listed that are of higher priority.

[0033] FIG. 6 shows the ME performing CS Authentication @ 30.868 s.

[0034] FIG. 7 shows the ME registered successfully on the CS domain of the non-preferred visited PLMN 302-880 @ 31.892 s, as indicted by the updated TMSI value in the EF_LOCI file and Location update status showing “updated”.

[0035] FIG. 8 shows the ME indicating to the UICC that it is now in Normal Service @ 31.930 s, which in this case is only on the CS domain, as PS registration has not yet occurred.

[0036] FIG. 9 shows the ME performing PS Authentication @ 32.358 s.

[0037] FIG. 10 shows the ME registered successfully on the PS domain of the non-preferred visited PLMN 302-880 @ 33.426 s, as indicted by the updated P-TMSI value in the EF_PSLOCI file and Routing Area update status showing “updated”. This Fig. can be referenced to Step 1 of FIG. 1, as the UE at this point is now fully registered on both the CS and PS domain on the non-preferred VPLMN.

[0038] FIG. 11 shows an SMS being sent to the ME @ 92.242 s. This SMS will update the EF_LOCI file on the UICC and instruct the UICC to send a REFRESH to the ME. This Fig. can be referenced to Step 2 of FIG. 1.

[0039] FIG. 12 shows the UICC sending REFRESH—3G Session Reset to the ME @ 97.453 s. The file path for

EF_LOCI is shown in the REFRESH File List, which indicates that EF_LOCI needs to be refreshed. The REFRESH shown in the COMPRIION IT3 log, can be used as the starting point to time how long it takes for the UE to register onto the preferred visited PLMN. This Fig. can be referenced to Step 4 of FIG. 1.

[0040] FIG. 13 shows the EF_LOCI being re-read by the ME @ 101.085 s, as part of the USIM Initialization process specified by REFRESH—3G Session Reset (Step 6 in FIG. 1) following a restart (Step 5 in FIG. 1). As shown, the MCC-MNC in the EF_LOCI has been updated with the preferred visited network PLMN ID 302-221. TMSI value and Location update status remains the same, which is key in order to achieve Instant Registration. This Fig. can be referenced to Step 7 of FIG. 1.

[0041] FIG. 14 shows the EF_PSLOCI being re-read by the ME @ 101.241 s, again as part of the USIM Initialization process specified by REFRESH—3G Session Reset. As shown, the EF_PSLOCI has not been updated and the MCC-MNC, P-TMSI value and Routing Area update status remains the same. However, it is not required to update the EF_PSLOCI, as registration on the CS domain via MCC-MNC update in EF_LOCI, will force a PS registration if allowed by the visited network. This Fig. can be referenced to Step 7 of FIG. 1.

[0042] FIG. 15 shows that the ME has performed the REFRESH—3G Session Reset successfully, as depicted by the TERMINAL RESPONSE @ 102.922 s

[0043] FIG. 16 shows the ME performing CS Authentication @ 105.545 s. This Fig. can be referenced to Step 8 of FIG. 1.

[0044] FIG. 17 shows the ME performing PS Authentication @ 105.872 s.

[0045] FIG. 18 shows the ME registered successfully on the PS domain of the preferred visited PLMN 302-221 @ 107.303 s, as indicated by the updated P-TMSI value in the EF_PSLOCI file and Routing Area update status showing “updated.”

[0046] FIG. 19 shows the ME registered successfully on the CS domain of the preferred visited PLMN 302-221 @ 107.503 s, as indicated by the updated TMSI value in the EF_LOCI file and Location update status showing “updated.”

[0047] FIG. 20 shows the ME fully registered on both CS and PS domain of the preferred visited PLMN 310-221 @ 107.534 s, as depicted by the EVENT DOWNLOAD—Normal Service sent from the ME to the UICC. This shows that the UE registered immediately after the REFRESH onto the preferred visited PLMN in 10.081 s. This Fig. can be referenced in Step 9 of FIG. 1.

[0048] FIG. 21 shows the interaction between the UICC and ME. Step 1 shows the ME registered on non-preferred 302880 PLMN as read from the LOCI file in the UICC. Step 2A shows the MCCMNC field updated in EF_LOCI, and Step 2B shows REFRESH 3G Session Reset then sent to the ME, and Step 2C shows the ME going into No Service briefly as it performs the REFRESH. Step 3 shows the ME registered on the preferred 302221 PLMN, and all values in EF_LOCI are updated.

[0049] ABBREVIATIONS used herein:

- [0050] 3GPP 3rd Generation Partnership Program
- [0051] ADF Application Dedicated File
- [0052] CS Circuit switched
- [0053] EF Elementary File
- [0054] GUI Graphic User Interface

- [0055] HPLMN Home Public Land Mobile Network
- [0056] HPPLMN Higher Priority Public Land Mobile Network
- [0057] IMSI International Mobile Subscriber Identity
- [0058] LAC Location Area Code
- [0059] LOCI Location Information
- [0060] ME Mobile Equipment
- [0061] MCC Mobile Country Code
- [0062] MM Mobility Management
- [0063] MNC Mobile Network Code
- [0064] MS Mobile Station
- [0065] OPLMNwAcT Operator Public Land Mobile Network with Access Technology
- [0066] OTA Over the Air
- [0067] PLMN Public Land Mobile Network
- [0068] PS Packet switched
- [0069] PSLOCI Packet switched Location Information
- [0070] P-TMSI Packet Temporary Mobile Subscriber Identity
- [0071] SIM Subscriber Identity Module
- [0072] SMS Short Message Service
- [0073] TMSI Temporary Mobile Subscriber Identity
- [0074] TR Technical Recommendation
- [0075] TS Technical Specification
- [0076] UE User Equipment
- [0077] UI User Interface
- [0078] UICC Universal Integrated Circuit Card
- [0079] USIM Universal Subscriber Identity Module
- [0080] VPLMN Visited Public Land Mobile Network

[0081] Immaterial modifications may be made to the embodiments described here without departing from what is covered by the claims. In the claims, the word “comprising” is used in its inclusive sense and does not exclude other elements being present. The indefinite article “a” before a claim feature does not exclude more than one of the feature being present. Each one of the individual features described here may be used in one or more embodiments and is not, by virtue only of being described here, to be construed as essential to all embodiments as defined by the claims.

1. A method of forcing a UE that is in Automatic Network Selection mode and is registered on a non-preferred visited network to register onto an available preferred visited network by the steps comprising overwriting the MCCMNC content of EF_LOCI in the USIMADF within the UICC, with the PLMN ID of an available preferred visited network, and then sending to the ME REFRESH—3G Session Reset containing EF_LOCI in the REFRESH File List.

2. The method of claim 1, where registration of the UE onto the available preferred visited network, is accomplished within 30 seconds taken from the time that the REFRESH—3G Session Reset is sent to the ME.

3. The method of claim 1, accomplished seamlessly, in which the ME does not power-cycle itself after REFRESH—3G Session Reset is sent to the ME.

4. The method of claim 1, where the ME complies to all of the following 3GPP requirements:

- a. 3GPP TS 31.111 clause 6.4.7—REFRESH
- b. 3GPP TS 23.122 clause 4.8—MM Restart procedures
- c. 3GPP TS 23.122 clause 4.4.3.1—At switch on or recovery from lack of coverage

5. The method of claim 1, in which there is a valid TMSI value available in EF_LOCI in the UE.

6. The method of claim 5 in which, in the UE, bytes 1 to 4 of File ID 6F7E have not been overwritten with FF FF FF FF.

7. The method of claim 1, in which, in the UE, the Location update status in EF_LOCI is set to “updated”.

8. The method of claim 7, in which, in the UE, byte 11 of File ID 6F7E is set to 00.

9. The method of claim 1, further comprising updating the EF_OPLMNwAcT with the preferred PLMN ID and Access Technology of the preferred visited network.

10. The method of claim 9 in which preferred PLMN ID is updated as the highest priority entry in the EF_OPLMNwAcT.

11. The method of claim 8, further comprising, after updating the EF_OPLMNwAcT, immediately sending REFRESH—3G Session Reset (containing EF_LOCI and EF_OPLMNwAcT in the REFRESH File List) to the ME.

12. The method of claim 1 carried out using a commercial OTA solution.

13. The method of claim 1 carried out using a SIM Toolkit application.

14. The method of claim 1 in which the MCCMNC content of EF_LOCI comprises bytes 5, 6 and 7 of File ID 6F7E.

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